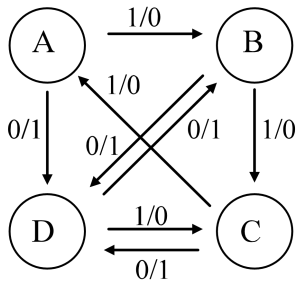
3. 请根据图 P5.1 所示的状态表画出相应的状态图，其中 X 为外部输入信号，Z 为外部输出信号，A、B、C、D 是时序电路的 4 种状态。

A/Z
X/出

Q^{n+1}/Z \ X, Q^n	0	1
A	D/1	B/0
B	D/1	C/0
C	D/1	A/0
D	B/1	C/0

图 P5.1



注意，此为米里型电路，输出在线上

4. 已知主从 RS 触发器的逻辑符号和 Clk、S、R 端的波形如图 P5.2（a）（b）所示，试画出 Q 端对应的波形（设触发器的初始状态为 0）。

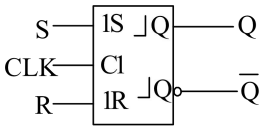


图 P5. 2 （a）

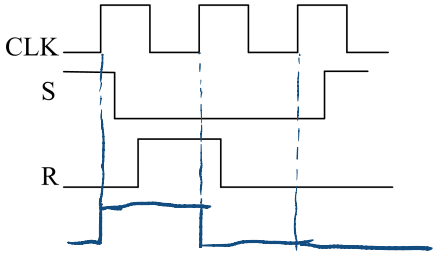
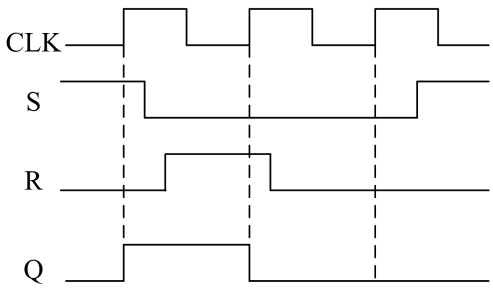


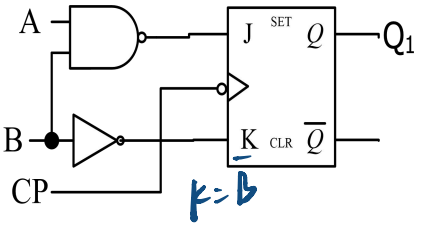
图 P5. 2（b）



6. 边沿型 JK 触发器（~~上升沿~~下降沿有效）的触发信号和输入信号如图 P5.4（a）（b）所示。试画出 Q_1 端的输出波形（所有触发器的初态为 0）。

合时间序入

$J=\overline{AB}$



$K=\overline{B}$

图 P5.4 (a)

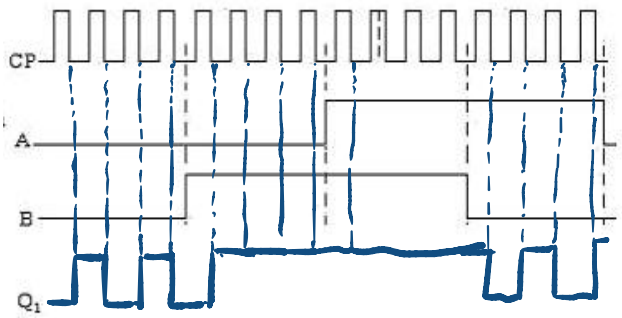
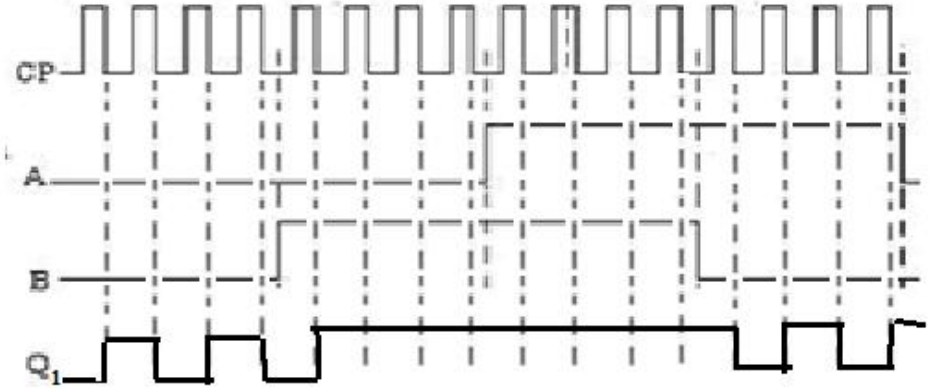


图 P5.4 (b)

$Q_1^{n+1}=\overline{AB}\,\overline{Q_1^n}+\overline{B}\cdot Q_1^n$

$\overline{Q_1^n}+\cancel{Q_1^n}$

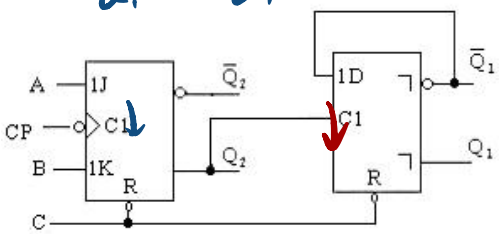
图 P5.4



7. 用主从 D 触发器和边沿触发的 JK 触发器组成的电路如图 P5.5（a）所示。已知触发信号和输入信号，如图 P5.5（b）所示。试画出 Q_1、Q_2 的输出波形（所有触发器的初态为 0）。

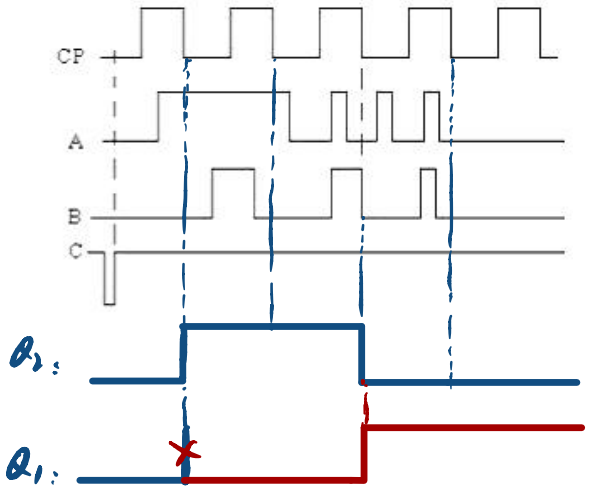
$Q_2^{n+1}=A\overline{Q_2^n}+\overline{B}Q_2^n$

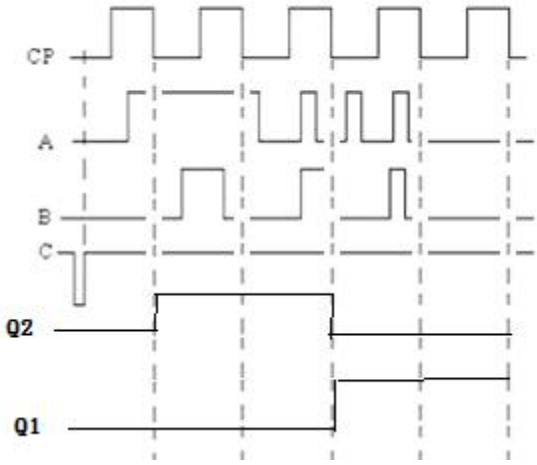
$Q_1^{n+1}=\overline{Q_1^n}$



主从D触发器为下降沿

图 P5.5





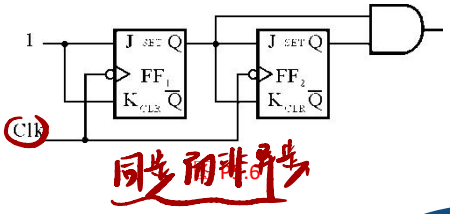
8. 分别用方程式、状态转换图表示如图 P5.6 所示电路的功能。

$J_1=1 \quad K_1=1$

$Q_1^{n+1}=\overline{Q_1^n}$

$J_2=Q_1^n=K_2 \quad Q_2^{n+1}=Q_1^n\overline{Q_2^n}+\overline{Q_1^n}Q_2^n$

$Z=Q_1^n\cdot Q_2^n$



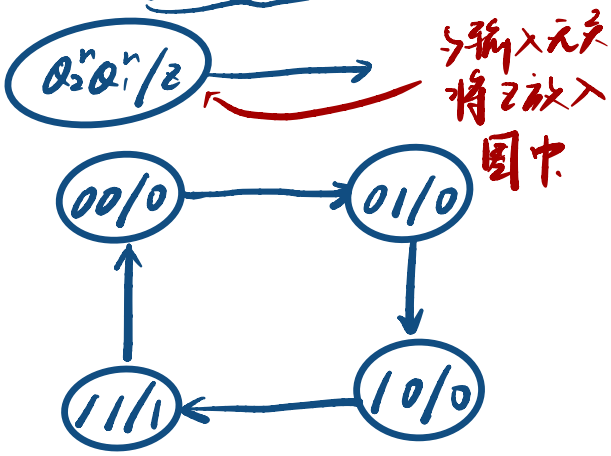
同步的非异步

$$Q_1^{n+1}=J_1\overline{Q_1^n}+\overline{K}_1Q_1^n=\overline{Q_1^n}$$

方程式：$J_2=K_2=Q_1^n$（千万不能成Q_1^{n+1}）

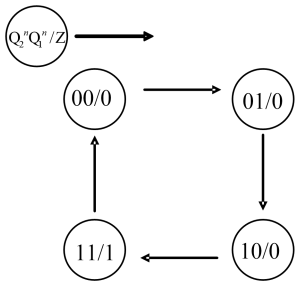
$$Q_2^{n+1}=J_2\overline{Q_2^n}+\overline{K_2}Q_2^n=Q_1^n\overline{Q_2^n}+\overline{Q_1^n}Q_2^n=Q_1^n\oplus Q_2^n$$

输出设为 Z：$Z=Q_1^n\cdot Q_2^n$（不能成$Q_1^{n+1}\cdot Q_2^{n+1}$）



→输入无关
将Z放入
图中

功能：同步的模4计数器，



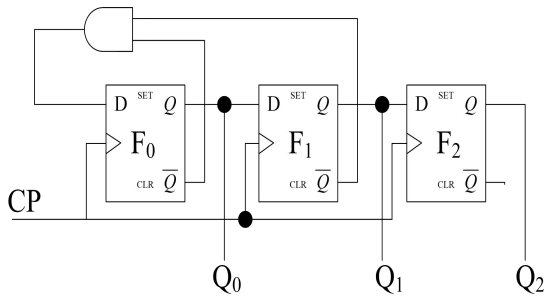
状态转换图：

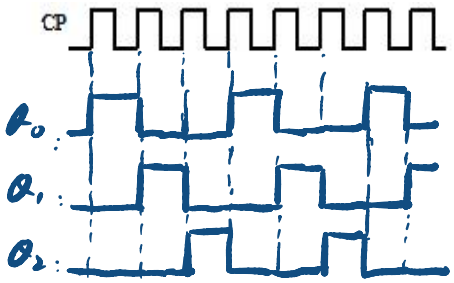
注意：输出在圈内而不是在线上！！

功能：由状态图可知，此电路为同步模 4 计数器，输出 Z 为最大值指示器；

或者此电路为输出 1000 序列信号；或者此电路在连续四个脉冲后输出 1。

11. 分析图 P5.9（a）（b）所示电路，画出状态转换图和时序波形图。





同步

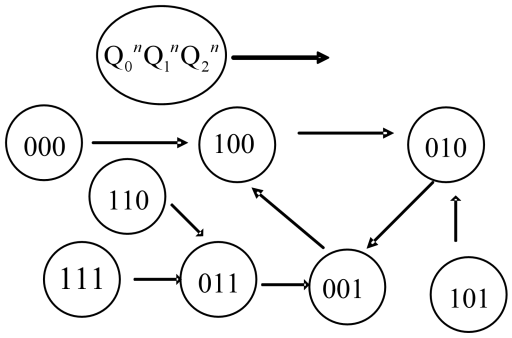
$$\begin{cases} Q_0^{n+1} = \overline{Q_0^n} \cdot \overline{Q_1^n} \\ Q_1^{n+1} = Q_0^n \\ Q_2^{n+1} = Q_1^n \end{cases}$$

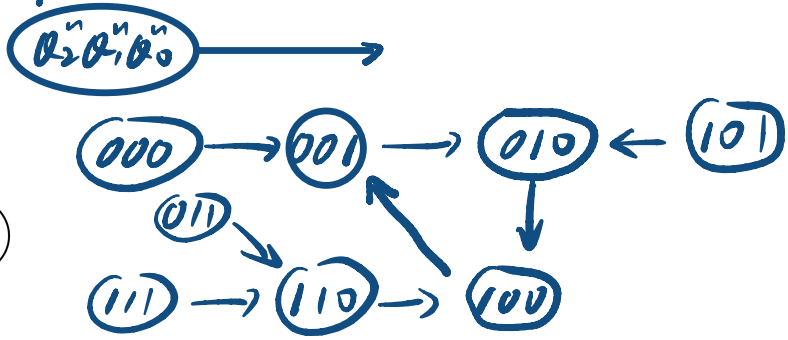
图 P5.9

$$Q_0^{n+1} = \overline{Q_0^n} \cdot \overline{Q_1^n}$$
$$Q_1^{n+1} = Q_0^n$$
$$Q_2^{n+1} = Q_1^n$$

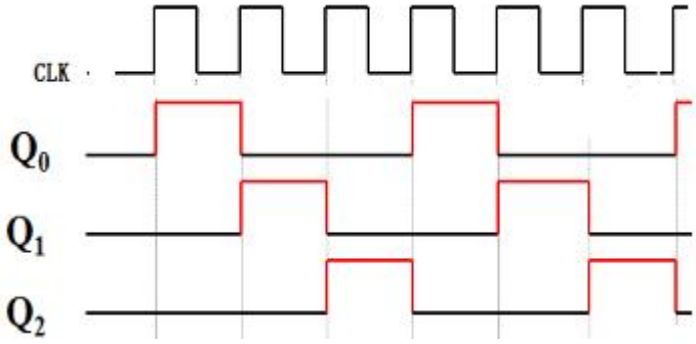
状态转换图：（不要少状态）

转移图：



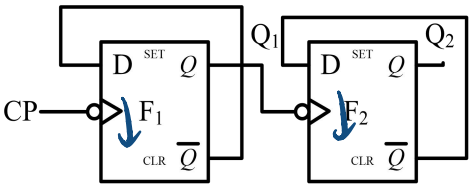


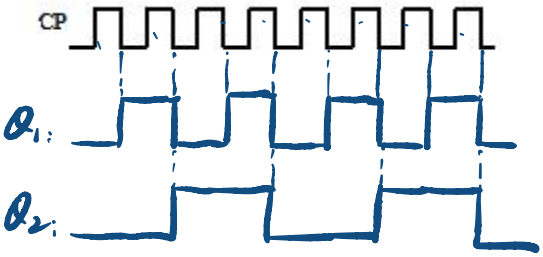
保证检查到每个点



波形图：

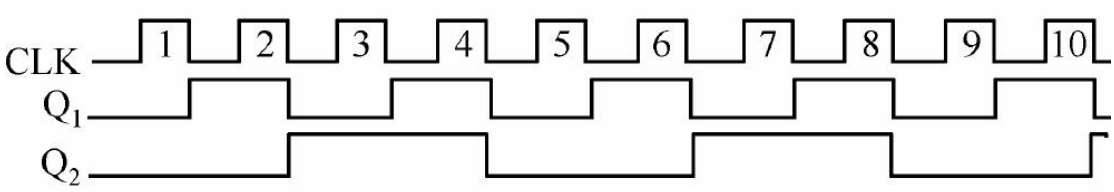
12. 逻辑电路如图 P5.10（a）（b）所示，对于连续脉冲，试画出 Q_1、Q_2 的波形。





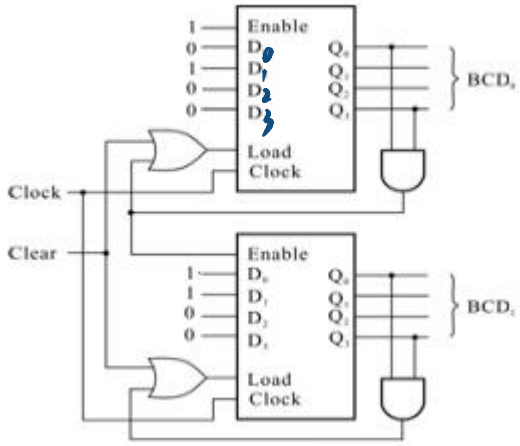
$$Q_1^{n+1} = \overline{Q_1^n}$$
$$Q_2^{n+1} = \overline{Q_2^n}$$

图 P5.10



13. 将图 5.36 中的“一个两位数的 BCD 计数器”改造为模 56 的计数器。

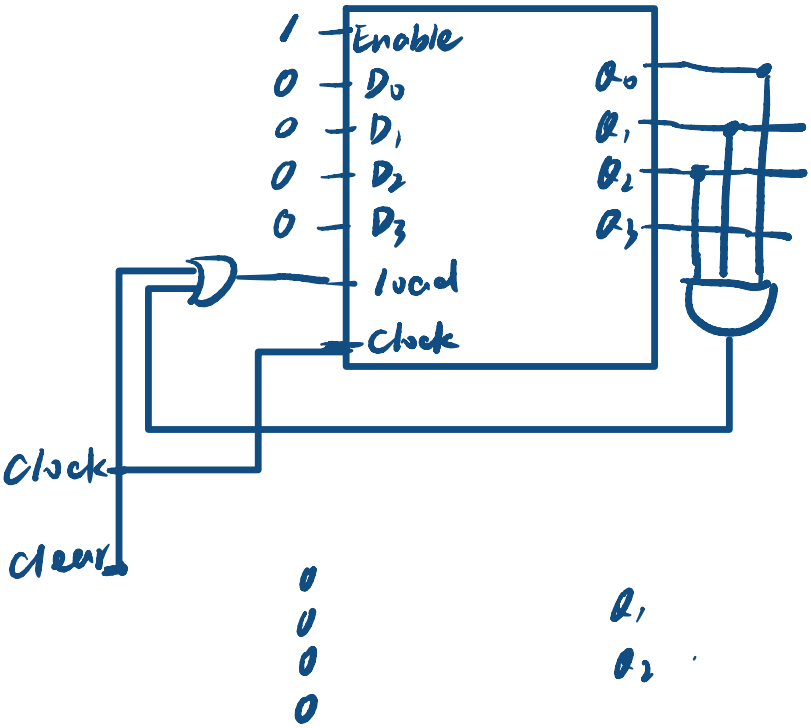
0010 1001
2 → 9
8
0011 → 1001
3 9
7

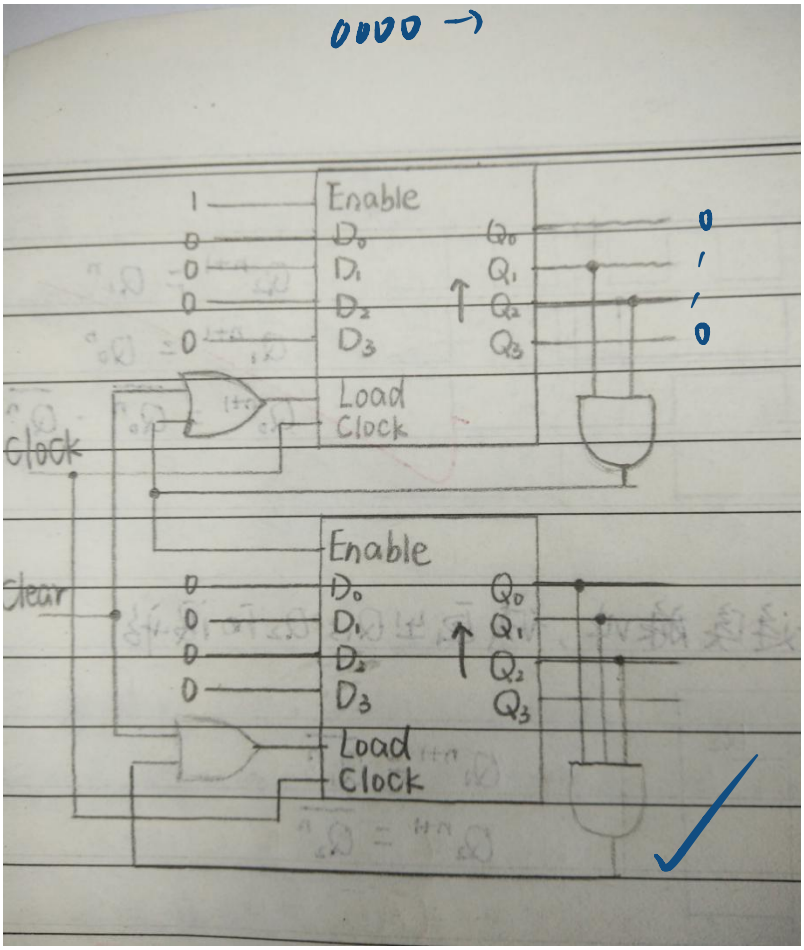


也可以按十进制数，从 00-60 到 99 再回到 44-99，可以认为是模 56 的计数器。此题可有多种方法。

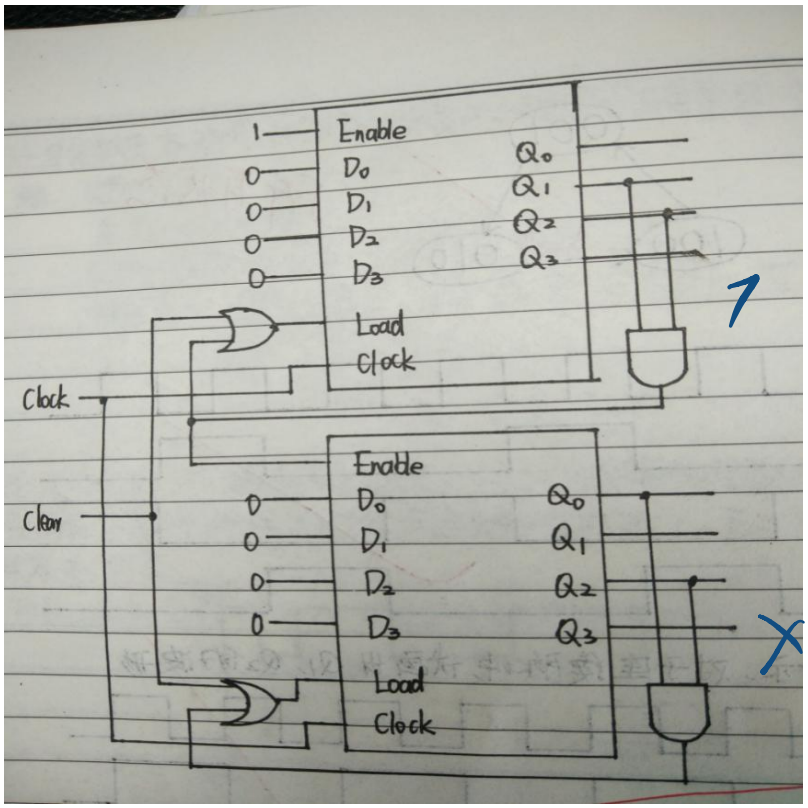
0 → 7.
0111

试判断以下方法是否可行？？

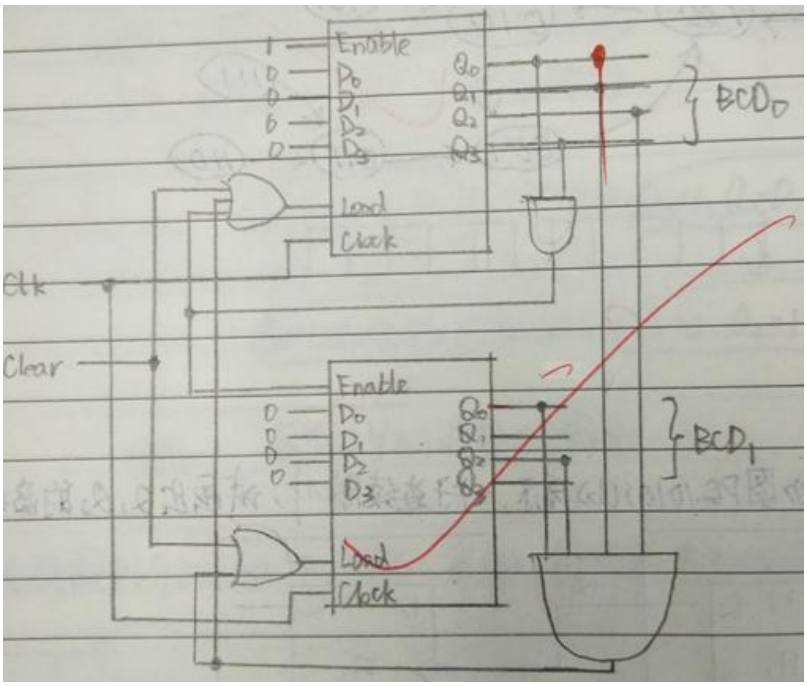




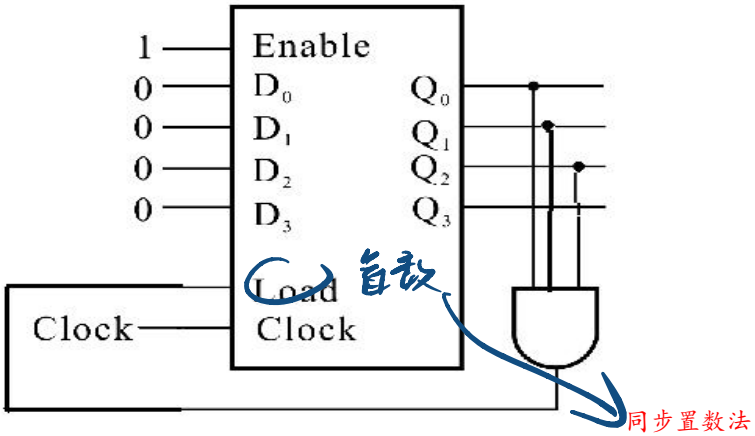
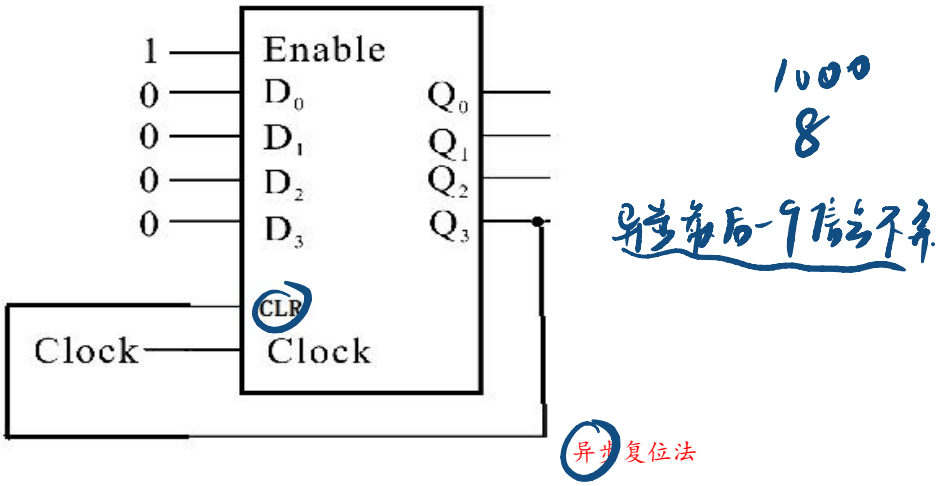
这个是对的，



这个是不对的，



这个也是不对的

注意此张图当满足 55 时十位的 Enable 信号还无效，所以不能满足置数，请考虑一下怎么改？

14. 针对一个模 16 的二进制计数器，分别用同步置数法和异步复位法构造一个模 8 计数器。